

To-Do List Manager

FastAPI tabanlı küçük bir görev yönetimi servisi üzerinde test, dağıtım, otomasyon ve gözlemlenebilirlik hattı kuruldu.

FastAPI

Docker

Kubernetes

LocalStack

Grafana

Playwright

20 passed

Test

%89

Coverage

31.44 ms

k6 p95

Başarılı

CI/CD

Proje amacı ve kapsam

- Basit bir To-Do List Manager servisi ile uçtan uca test mühendisliği pratiği gösterildi.
- Task ve Tag entity'leri üzerinden görev oluşturma, listeleme, okuma, tamamlama, silme, etiketleme, dosya yükleme ve istatistik akışları sağlandı.
- Odak, karmaşık uygulama yerine ölçülebilir kalite, otomasyon ve teslimat altyapısı kurmaktır.

Tek ekran görev yönetimi

To-Do List Manager

Secure cloud delivery task board for test engineering demos

Demo Workspace

FastAPI 0.2.0

API ONLINE

NEW TASK

SEC-OPS

Title

Patch exposed endpoint

Description

Add tests, verify metrics, close risk

Tag

incident, ci, k8s, s3

Ekle

SESSION

Operator

Environment

Evidence

Demo User

LocalStack

Metrics enabled

Presentation checklist

- API endpoints visible in Swagger
- Prometheus metrics exposed
- Tasks, tags, S3 attachments ready

OPERATIONS BOARD

25% CLOSED

4

Total tasks

3

Open

1

Closed

4

Visible

SEARCH

Search tasks or tags

All

Open

Closed

Check Grafana dashboard

Check Grafana dashboard for monitoring

#monitoring

OPEN

Tamamla

Dosya

Sil

Run k6 latency scenario

#performance

OPEN

Tamamla

Dosya

Sil

Validate LocalStack S3 upload

#s3

OPEN

Tamamla

Dosya

Sil

Write unit tests for task API

Applied

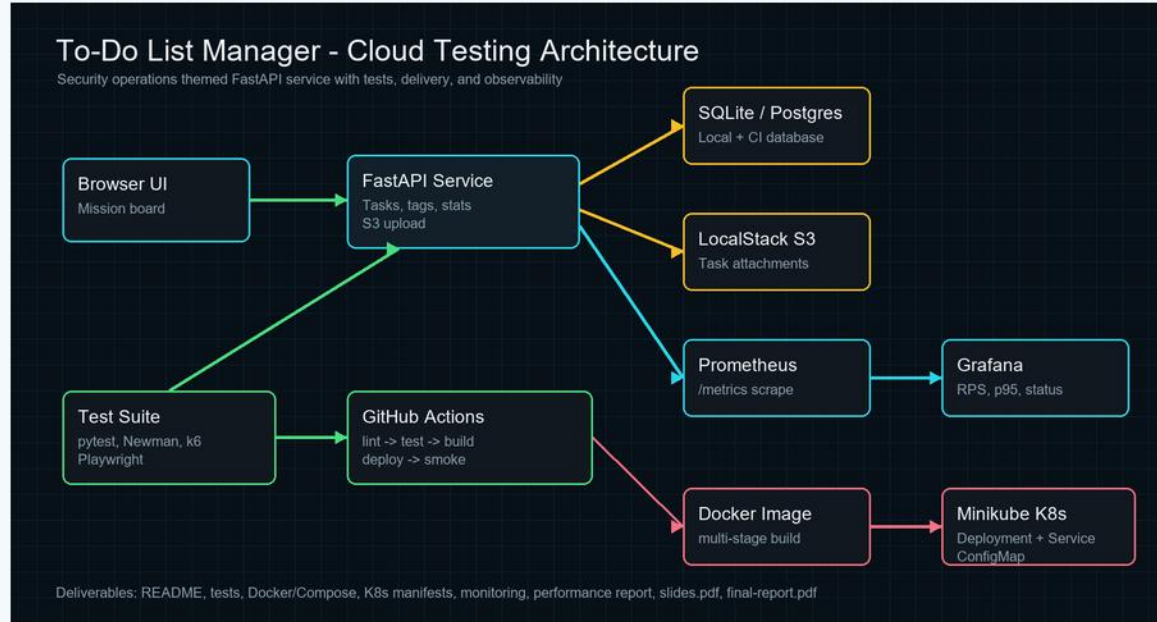
CLOSED

Dosya

Sil

Arama, durum filtresi, etiketler, görev aksiyonları ve API çevrim içi bilgisi aynı ekranda.

Mimari bileşenler birlikte çalışır



FastAPI, PostgreSQL/SQLite, LocalStack S3, Prometheus, Grafana, Docker, Kubernetes ve GitHub Actions.

Test stratejisi şartnamedeki %70 coverage eşiğini aştı

```
(base) PS C:\Users\mdapc\OneDrive\Masaüstü\bulutproje\bulut-test-project> pytest --cov=src --cov-report=term-missing
===== test session starts =====
platform win32 -- Python 3.11.15, pytest-9.0.3, pluggy-1.6.0
rootdir: C:\Users\mdapc\OneDrive\Masaüstü\bulutproje\bulut-test-project
configfile: pyproject.toml
testpaths: tests
plugins: anyio-4.12.1, Faker-40.19.1, base-url-2.1.0, cov-7.1.0, playwright-0.8.0
collected 20 items

tests\e2e\test_ui.py ... [ 15%]
tests\integration\test_db.py ... [ 30%]
tests\unit\test_s3.py .... [ 50%]
tests\unit\test_tasks.py ..... [100%]

===== warnings summary =====
..\..\..\miniconda3\Lib\site-packages\fastapi\testclient.py:1
C:\Users\mdapc\miniconda3\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient' is deprecated; install
'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa

tests\e2e\test_ui.py::test_homepage_loads[chromium]
C:\Users\mdapc\miniconda3\Lib\site-packages\websockets\legacy\__init__.py:6: DeprecationWarning: websockets.legacy is deprecated; see https://websockets.readthedoc
s.io/en/stable/howto/upgrade.html for upgrade instructions
  warnings.warn( # deprecated in 14.0 - 2024-11-09

tests\e2e\test_ui.py::test_homepage_loads[chromium]
C:\Users\mdapc\miniconda3\Lib\site-packages\uvicorn\protocols\websockets\websockets_impl.py:17: DeprecationWarning: websockets.server.WebSocketServerProtocol is de
precated
  from websockets.server import WebSocketServerProtocol

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== tests coverage =====
coverage: platform win32, python 3.11.15-final-0

Name           Stmts   Miss  Cover   Missing
-----
src\__init__.py      0      0   100%
src\main.py         102    16    84%   68-69, 145, 160-178
src\models.py        17      0   100%
src\services\__init__.py  0      0   100%
src\services\s3.py   22      0   100%
TOTAL              141    16    89%

===== 20 passed, 3 warnings in 8.65s =====
```

pytest --cov=src sonucu: 20 passed, %89 coverage

Unit

Task API ve S3 servis mantığı

Integration

PostgreSQL Testcontainers ile DB doğrulama

E2E

Playwright ile UI senaryoları

Coverage

Toplam %89, hedefin üzerinde

CI/CD kalite kapılarını otomatik çalıştırıyor

test-and-build

succeeded 8 hours ago in 2m 13s

Search logs



> Set up job	1s
> Initialize containers	20s
> Checkout code	1s
> Set up Python	0s
> Install dependencies	14s
> Run Lint	1s
> Install Playwright Browsers	22s
> Run Pytest & Coverage	12s
> Setup Node.js	5s
> Install Newman	6s
> Run Newman Tests & Smoke Test	23s
> Docker Build	24s
> Post Setup Node.js	0s
> Post Set up Python	0s
> Post Checkout code	1s
> Stop containers	0s
> Complete job	0s

Black lint, pytest coverage, Playwright kurulumu, Newman smoke test ve Docker build adımları başarılı.

Gözlemlenebilirlik ve performans kanıtları



```

(base) PS C:\Users\bulut\OneDrive\Masain\bulut\projeler\bulut-test-project> k6 run perf/load-test.js

      Grafana
      MR6

execution: local
script: perf/load-test.js
output:

scenarios: (100.00%) 1 scenario, 10 max VUs, 10s max duration (incl. graceful stop):
  * default: 10 looping VUs for 30s (gracefulStop: 30s)

THRESHOLDS
  http_req_duration
    / p{95}<500 p{99}<11.4ms

TOTAL RESULTS
checks_total..... 300 100.00%
checks_succeeded... 100.00% 300 out of 300
checks_failed..... 0.00% 0 out of 300
/ in status 300

HTTP
http_req_duration..... avg:13.9ms min:0.29ms med:7.88ms max:150.7ms p{50}:11.1ms p{95}:11.4ms
[expected_response:true] avg:13.9ms min:0.29ms med:7.88ms max:150.7ms p{50}:11.1ms p{95}:11.4ms
http_req_failed..... 0.00% 0 out of 300
http_reqs..... 300 100.00%

EXECUTION
Iteration_duration..... avg:1.81s min:1s med:1s max:1.15s p{50}:1.82s p{95}:1.85s
Iterations..... 300 100.00%
VUs..... 10 avg:10s max:10s
VUs_max..... 10 min:10s max:10s

NETWORK
data_received..... 107 KB 15.65%
data_sent..... 21 KB 3.08%

running (0m0.5s), 00/10 VUs, 300 complete and 0 interrupted iterations
default [-----] 10 VUs 30s
  
```

- Request rate
- P95 latency
- HTTP status code
- k6 p95: 31.44 ms

SONUÇ

Sonuç: servis, test ve dağıtım hattı birlikte savunulabilir durumda

```
(base) PS C:\Users\mdapc\OneDrive\Masaüstü\bulutproje\bulut-test-project> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
todo-list-manager-67d8c764f7-lp9rv  1/1     Running   0           28m
todo-list-manager-67d8c764f7-txtmk  1/1     Running   0           28m
(base) PS C:\Users\mdapc\OneDrive\Masaüstü\bulutproje\bulut-test-project> kubectl get svc
NAME                                TYPE               CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes                         ClusterIP          10.96.0.1    <none>        443/TCP          29m
todo-list-manager-service          NodePort           10.99.192.88 <none>        80:30080/TCP     29m
```

20 test geçti

%89 coverage

Başarılı CI run

Kubernetes podları Running

p95 latency 31.44 ms

Proje, şartnamedeki ana teslimleri küçük ama açık biçimde savunulabilir bir mikroservis üzerinde birleştirir.